Muhammad Zahid

Senior Data Engineer

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SUMMARY

Data engineer with 4 years of experience owning production data pipelines end to end — ingestion, transformation, enrichment, validation, and monitoring — for high-volume, real-time systems. Deep expertise in Kafka, ClickHouse, Kubernetes, and AWS, with hands-on experience debugging across orchestration, storage, and warehouse layers, designing schema evolution and data-contract guardrails, and integrating ML/NLP models into production pipelines.

SKILLS

- **Languages:** Python, SQL, Go (working), Rust (working), JavaScript, Bash, C/C++
- **Orchestration:** Apache Airflow, Mage AI, Kafka, PySpark, Spark Structured Streaming, Debezium CDC, dbt
- **Data Platforms:** ClickHouse, Snowflake-adjacent (Trino), PostgreSQL, MongoDB, Redis, Delta Lake, Redshift, MySQL, MS SQL Server
- **Infra & Orchestration Systems:** Kubernetes, Docker, KEDA, AWS (EC2, S3, SQS, Lambda, Glue, EKS), Azure (Databricks, DevOps), Terraform (working)
- **Data Modeling & Quality:** Star/Snowflake schemas, SCD Type 2/4, schema evolution, data cleaning, data mart design, entity normalization
- **Observability:** Grafana, query tuning (EXPLAIN), CI/CD (GitHub Actions, Azure DevOps), GitOps

PROFESSIONAL EXPERIENCE

Senior Data Engineer, Salla 09/2025 | Remote, Finland

- Own end-to-end ELT pipelines integrating ERP, transactional, and event data into ClickHouse for near-real-time analytics, deployed on AWS EKS with Terraform-managed infrastructure
- Designed and operated event-driven microservices (Python/JavaScript) consuming Kafka streams with Redis integration; containerized with Docker and orchestrated on Kubernetes with KEDA-driven elastic scaling tied to Kafka consumer lag/backlog, improving cost efficiency under variable traffic
- Diagnosed and resolved a Kafka consumer bottleneck, achieving a 1000x throughput improvement (25 → 25,000 msg/s)
- Implemented forward-compatible schema evolution for ClickHouse tables: new upstream columns are captured into a structured *additional*columns JSON field rather than breaking downstream consumers, avoiding blocking schema migrations
- Owned end-to-end ClickHouse administration — schema design (ORDER BY, partitioning, skip indexes), query tuning, and 2-replica high availability — reducing query latency significantly
- Built a multi-threaded ClickHouse backup/restore framework to S3 + Iceberg (hive-style partitioning, PostgreSQL metadata) with Slack-based monitoring/alerting
- Developed an event-driven NetSuite ERP sync pipeline (SQS + Lambda + dbt) for financial/invoicing data with full reconciliation
- Built an S3 file-combiner microservice in Rust to reduce small-file overhead in the storage layer
- Migrated orchestration from Airflow to Mage AI, improving reliability and SLA adherence
- Partnered with the DevOps team on EKS cost optimization, tuning instance/scaling strategy against real-time Kafka backlog metrics

Data Engineer, Addo AI 04/2024 – 09/2025 | Remote, Finland

- Led architecture of a government analytics platform processing terabytes of multi-sectoral data daily (~200 Airflow DAGs), designing an on-prem/cloud lakehouse on Azure Databricks, Delta Lake, Trino, and ClickHouse
- Migrated a legacy Oracle 19c system to AWS using Glue jobs, with Redshift as the target warehouse
- Built a generic ETL framework cutting deployment time by 60% and processing time from 10 to 5 hours, supporting ingestion from MySQL, SQL Server, APIs, web scraping, and PDFs
- Implemented real-time streaming (Kafka + PySpark) and dimensional models (fact/dimension tables, SCD Type 2/4) with geographic normalization for entity/location matching across sources
- Integrated an NLP-based name-normalization model into the production pipeline to correctly parse and standardize Korean names, improving downstream entity-matching accuracy
- Led ClickHouse adoption as the OLAP engine, delivering a 40% query performance gain; automated RBAC and anonymization, cutting manual effort by 90%
- Acted as primary technical point of contact with client teams, debugging data correctness issues and discrepancies directly with stakeholders
- Led a team of 3 engineers delivering BI dashboards and Python-based ingestion automation (API, GeoJSON, DAG orchestration)

Data Engineer, RedCoast Co 08/2023 – 03/2024 | Ontario, Canada

- Built ETL pipelines in Python + Airflow for multiple clients; implemented Spark Structured Streaming for low-latency processing, deployed on Azure

- Developed a dynamic SQL Builder enabling business users to generate custom KPIs without writing SQL; integrated MLflow for model versioning
- Built Python scripts and backend services for data ingestion and analytics across PostgreSQL, MongoDB, and other sources

Software Engineer, Bitnine Global 08/2022 – 08/2023 | Seoul, South Korea

- **Open Source Contribution:** Developed core features (inheritance, mathematical functions, string manipulation) for **Apache AGE**, an open-source graph database extension for PostgreSQL, and contributed to **PG-Pool II**.
- **Database Software Development:** Built database applications using **C/C++ and Python**, leveraging Figma designs and Tkinter for desktop/web interfaces.
- **Multi-Database Expertise:** Worked across **PostgreSQL, Oracle, and AgensSQL**; served as main developer for the **AgensSQL Migration Tool**.

Cloud Computing Intern, Nokia 08/2022 – 09/2022 | Remote, Pakistan

Lean about Telco Cloud Computing infrastructure

EDUCATION

M.S.c Computer Science, Åbo Akademi University 08/2025 – Present | Turku, Finland

B.S.c Computer Science 08/2019 – 06/2023 | Islamabad, Pakistan
National University of Computer and Emerging Sciences (FAST NUCES)

PROJECTS

OSS Online Statistical System (PITB) - Airflow, Trino, Delta Lake, Kafka, ClickHouse, Superset

- Led a team of 6+ engineers to build a fully open-source, on-premises data lakehouse under strict data sovereignty constraints, ingesting terabytes of government data daily across Tax, Health, Education, and Finance sectors
- Designed an automated RBAC governance system and delivered multi-sector BI dashboards, cutting anonymisation effort by 90% and reducing pipeline delays by 50%

Streaming Data Pipeline - PySpark Structured Streaming, Kafka, ClickHouse

- Built a real-time pipeline with windowed aggregations and multi-stream joins to derive in-flight business metrics, persisted to ClickHouse with a schema optimized for low-latency OLAP querying

NetSuite ERP Data Sync Pipeline - dbt, SQL, AWS

Built a pipeline extracting financial and transactional e-commerce data, transforming it via SQL models covering **invoicing and reconciliation**, and syncing it to NetSuite ERP for financial reporting

Daily Sales Report Automation (Haleon) - Azure, Snowflake

Built ELT pipelines ingesting from Excel, REST APIs, and email with incremental loading into Snowflake and data quality checks; delivered a Power BI dashboard validating the data model with the commercial team

Urdu Text-to-Speech Android App

Integrated a custom voice-cloning TTS model into a **React Native** Android app, handling UI state, async model calls, and playbac

Worldwide Remote Job Aggregator - TypeScript, Next.js, SQLite/Drizzle, Playwright

Designed and built a multi-layer job-postings pipeline covering remote job boards, direct ATS polling (Greenhouse, Lever, Ashby, Workday, and 8+ others), and a company-discovery engine sourcing from search APIs.

Generalized Data Modeling Agent (in progress)

Automating the process of data modelling through multi-agent approach. Each agent has a set of instruction/subprocess it is responsible for. For now making it so that it is compatible with claude.